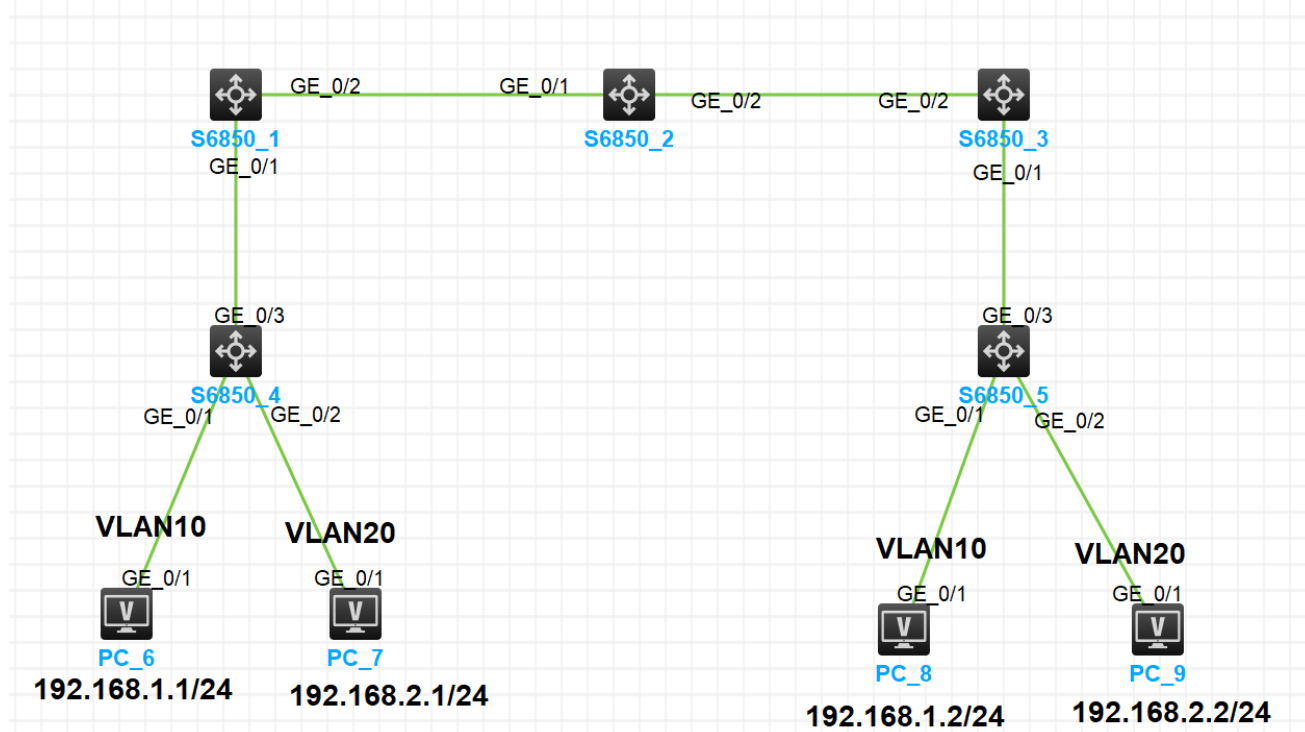


实验拓扑

1.按照图示配置 IP 地址

3.在 SW1 和 SW3 上通过 VLAN Mapping 方式配置灵活 QinQ，为收到的来自 VLAN 10 的帧打上 VLAN 100 的 Tag，来自 VLAN 20 的帧打上 VLAN 200 的 Tag，实现两端 VLAN 10 和 VLAN 20 的 PC 互通



实验需求

1. 按照图示配置 IP 地址
2. 按照图示在 SW4 和 SW5 上配置连接 PC 的接口分别加入 VLAN 10 和 VLAN 20
3. 在 SW1 和 SW3 上通过 VLAN Mapping 方式配置灵活 QinQ，为收到的来自 VLAN 10 的帧打上 VLAN 100 的 Tag，来自 VLAN 20 的帧打上 VLAN 200 的 Tag，实现两端 VLAN 10 和 VLAN 20 的 PC 互通

实验配置

```
[SW4]vlan 10
[SW4-vlan10]port GigabitEthernet 1/0/1
[SW4-vlan10]vlan 20
[SW4-vlan20]port GigabitEthernet 1/0/2
[SW4-vlan20]int g1/0/3

[SW4-GigabitEthernet1/0/3]port link-type trunk
```

```

[SW4-GigabitEthernet1/0/3]port trunk permit vlan 10 20
---
[SW5]vlan 10
[SW5-vlan10]port GigabitEthernet 1/0/1
[SW5-vlan10]vlan 20
[SW5-vlan20]port GigabitEthernet 1/0/2
[SW5-vlan20]int g1/0/3
[SW5-GigabitEthernet1/0/3]port link-type trunk
[SW5-GigabitEthernet1/0/3]port trunk permit vlan 10 20
#####
[SW1]vlan 10
[SW1-vlan10]vlan 20
[SW1-vlan20]vlan 100
[SW1-vlan100]vlan 200
[SW1-vlan200]int g1/0/1
[SW1-GigabitEthernet1/0/1]port link-type hybrid
[SW1-GigabitEthernet1/0/1]port hybrid vlan 10 20 tagged
[SW1-GigabitEthernet1/0/1]port hybrid vlan 100 200 untagged
[SW1-GigabitEthernet1/0/1]vlan mapping nest single 10 nested-vlan 100
[SW1-GigabitEthernet1/0/1]vlan mapping nest single 20 nested-vlan 200
[SW1-GigabitEthernet1/0/1]int g1/0/2
[SW1-GigabitEthernet1/0/2]port link-type trunk
[SW1-GigabitEthernet1/0/2]port trunk permit vlan 100 200
---
[SW3]vlan 10
[SW3-vlan10]vlan 20
[SW3-vlan20]vlan 100
[SW3-vlan100]vlan 200
[SW3-vlan200]int g1/0/1
[SW3-GigabitEthernet1/0/1]port link-type hybrid
[SW3-GigabitEthernet1/0/1]port hybrid vlan 10 20 tagged
[SW3-GigabitEthernet1/0/1]port hybrid vlan 100 200 untagged
[SW3-GigabitEthernet1/0/1]vlan mapping nest single 10 nested-vlan 100
[SW3-GigabitEthernet1/0/1]vlan mapping nest single 20 nested-vlan 200
[SW3-GigabitEthernet1/0/1]int g1/0/2
[SW3-GigabitEthernet1/0/2]port link-type trunk
[SW3-GigabitEthernet1/0/2]port trunk permit vlan 100 200
#####
[SW2]vlan 100
[SW2-vlan100]vlan 200
[SW2-vlan200]int ra g 1/0/1 g 1/0/2
[SW2-if-range]port link-type trunk
[SW2-if-range]port trunk permit vlan 100 200

```

结果验证

```

<H3C>ping 192.168.1.2
Ping 192.168.1.2 (192.168.1.2): 56 data bytes, press CTRL_C to break

```

```

56 bytes from 192.168.1.2: icmp_seq=0 ttl=255 time=4.222 ms
56 bytes from 192.168.1.2: icmp_seq=1 ttl=255 time=2.542 ms
---
<H3C>ping 192.168.2.2
Ping 192.168.2.2 (192.168.2.2): 56 data bytes, press CTRL_C to break
Request time out
Request time out
#####
<H3C>ping 192.168.1.2
Ping 192.168.1.2 (192.168.1.2): 56 data bytes, press CTRL_C to break
Request time out
Request time out
---
<H3C>ping 192.168.2.2
Ping 192.168.2.2 (192.168.2.2): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.2: icmp_seq=0 ttl=255 time=4.724 ms
56 bytes from 192.168.2.2: icmp_seq=1 ttl=255 time=2.257 ms

```

40	1304.51/129	ac:83:10:04:01:00	Spanning-tree- (TOR-...	119	MSL. ROOT = 32/00/0/ac:83:10:04:01:00	COST = 0
41	1304.702335	192.168.1.1	192.168.1.2	ICMP	106	Echo (ping) request id=0x00a3, seq=2/512, ttl=25
42	1304.703597	192.168.1.2	192.168.1.1	ICMP	106	Echo (ping) reply id=0x00a3, seq=2/512, ttl=25
43	1304.907500	192.168.1.1	192.168.1.2	ICMP	106	Echo (ping) request id=0x00a3, seq=3/768, ttl=25
44	1304.908649	192.168.1.2	192.168.1.1	ICMP	106	Echo (ping) reply id=0x00a3, seq=3/768, ttl=25
45	1305.112040	192.168.1.1	192.168.1.2	ICMP	106	Echo (ping) request id=0x00a3, seq=4/1024, ttl=2
46	1305.113209	192.168.1.2	192.168.1.1	ICMP	106	Echo (ping) reply id=0x00a3, seq=4/1024, ttl=2
47	1305.213810	ac:83:10:04:01:04	Broadcast	0xb003	22	Ethernet II
48	1306.224745	ac:83:11:00:02:04	Broadcast	0xb003	22	Ethernet II


```

> Frame 41: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0
> Ethernet II, Src: ac:83:30:f3:06:06 (ac:83:30:f3:06:06), Dst: ac:83:33:24:08:06 (ac:83:33:24:08:06)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
  000. .... = Priority: Best Effort (default) (0)
  ...0 .... = DEI: Ineligible
  .... 0000 0110 0100 = ID: 100
  Type: 802.1Q Virtual LAN (0x8100)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10
  000. .... = Priority: Best Effort (default) (0)
  ...0 .... = DEI: Ineligible
  .... 0000 0000 1010 = ID: 10
  Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.2
> Internet Control Message Protocol

```

198	1396.532435	192.168.2.1	192.168.2.2	ICMP	106 Echo (ping) request
199	1396.533727	192.168.2.2	192.168.2.1	ICMP	106 Echo (ping) reply
200	1396.623379	ac:83:10:04:01:00	Spanning-tree-(for-...	STP	119 MST. Root = 32768,
201	1397.307430	ac:83:10:04:01:04	Broadcast	0xb003	22 Ethernet II
202	1398.307470	ac:83:11:00:02:04	Broadcast	0xb003	22 Ethernet II

- > Frame 198: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0
- > Ethernet II, Src: ac:83:32:0c:07:06 (ac:83:32:0c:07:06), Dst: ac:83:34:31:09:06 (ac:83:34:31:09:06)
- 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
 - 000. = Priority: Best Effort (default) (0)
 - ...0 = DEI: Ineligible
 - 0000 1100 1000 = ID: 200
 - Type: 802.1Q Virtual LAN (0x8100)
 - 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
 - 000. = Priority: Best Effort (default) (0)
 - ...0 = DEI: Ineligible
 - 0000 0001 0100 = ID: 20
 - Type: IPv4 (0x0800)
 - > Internet Protocol Version 4, Src: 192.168.2.1, Dst: 192.168.2.2
 - > Internet Control Message Protocol